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Studierichting: Informatica

Oefeningenreeks 4A nr. 48.7

$$\begin{aligned} \int \frac{6}{e^{2x} + e^x - 2} dx &\stackrel{t=e^x}{=} 6 \int \frac{1}{t^2 + t + 2} \cdot \frac{1}{t} dt = 6 \int \frac{1}{t(t-1)(t+2)} dt \\ \Rightarrow \frac{1}{t(t-1)(t+2)} &= \frac{A}{t} + \frac{B}{t-1} + \frac{C}{t+2} \\ \Rightarrow A &= \frac{1}{-1 \cdot 2} = -\frac{1}{2} \\ \Rightarrow B &= \frac{1}{1(1+2)} = \frac{1}{3} \\ \Rightarrow C &= \frac{1}{-2(-2-1)} = \frac{1}{6} \\ &= -3 \int \frac{1}{t} dt + 2 \int \frac{1}{t-1} dt + \int \frac{1}{t+2} dt \\ &= -3 \ln |t| + 2 \ln |t-1| + \ln |t+2| + c \\ &\stackrel{t=e^x}{=} -3x + 2 \ln |e^x - 1| + \ln |e^x + 2| + c \end{aligned}$$

References